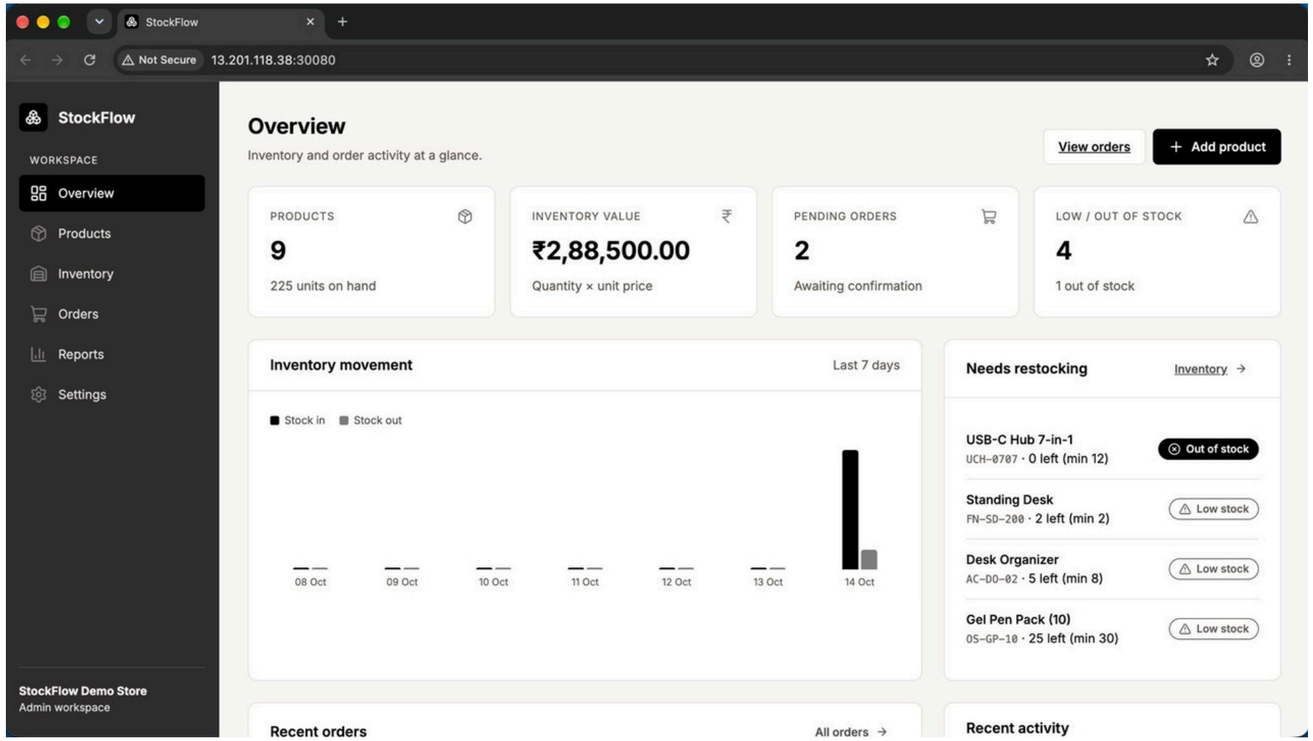
StockFlow: Inventory & Order Management on AWS EKS

DevOps Capstone · Sambhav D Bohra · Batch B · 24bcs10090

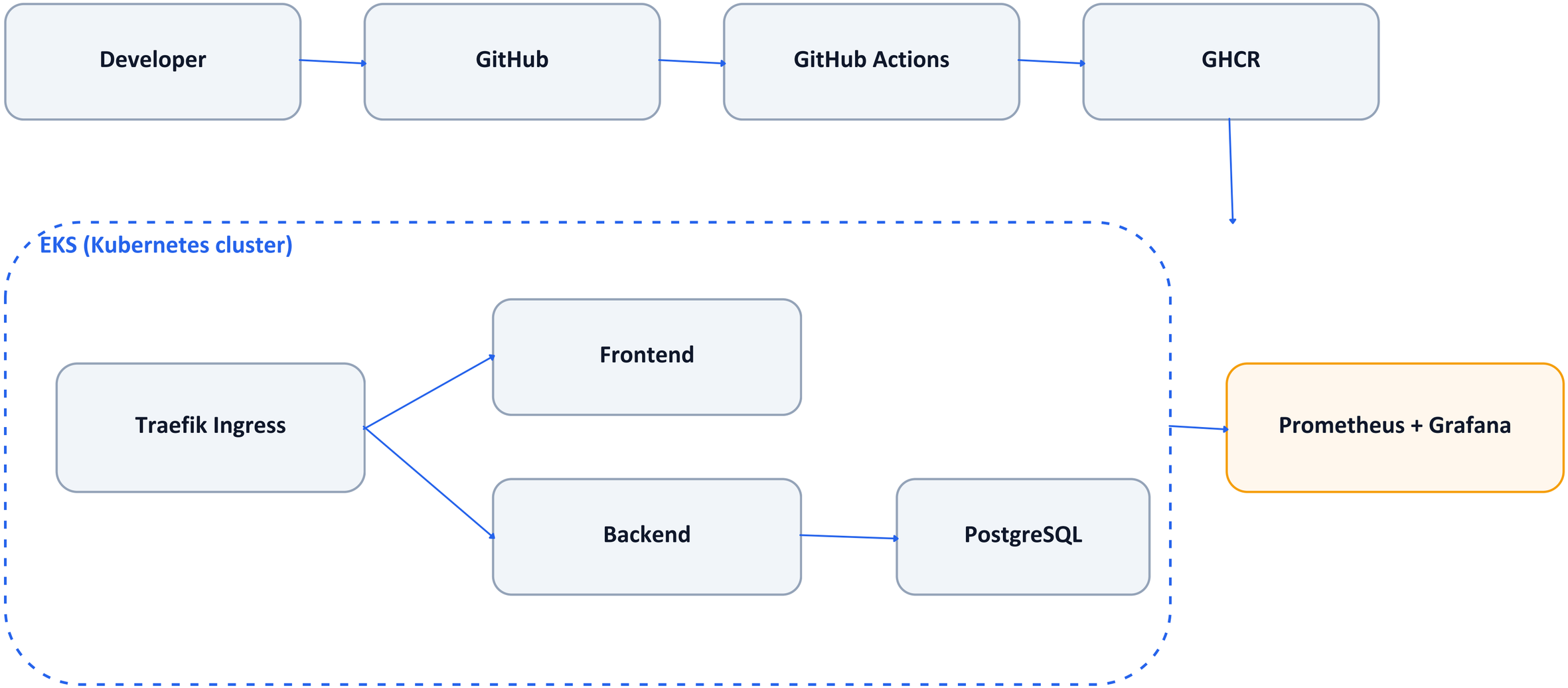
GitHub: github.com/sambhavbohra/StockFlow---DevOps-Project

The application

- Products, stock movements, orders (Pending → Confirmed → Shipped → Delivered / Cancelled), reports
- Stock deducted only on confirm; cancel restores it once; row locks for concurrency
- React + Vite · FastAPI · PostgreSQL · SQLAlchemy + Alembic · 41 pytest tests

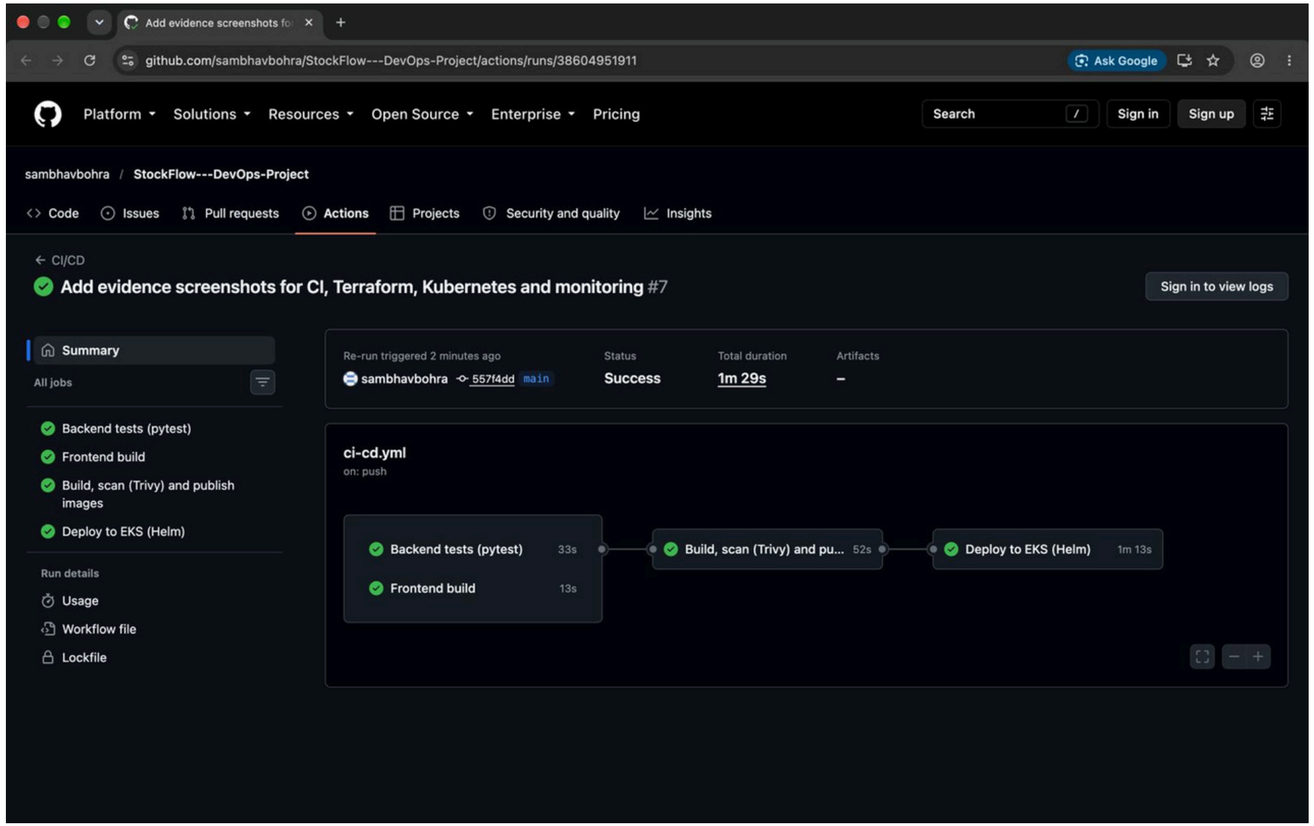


Architecture



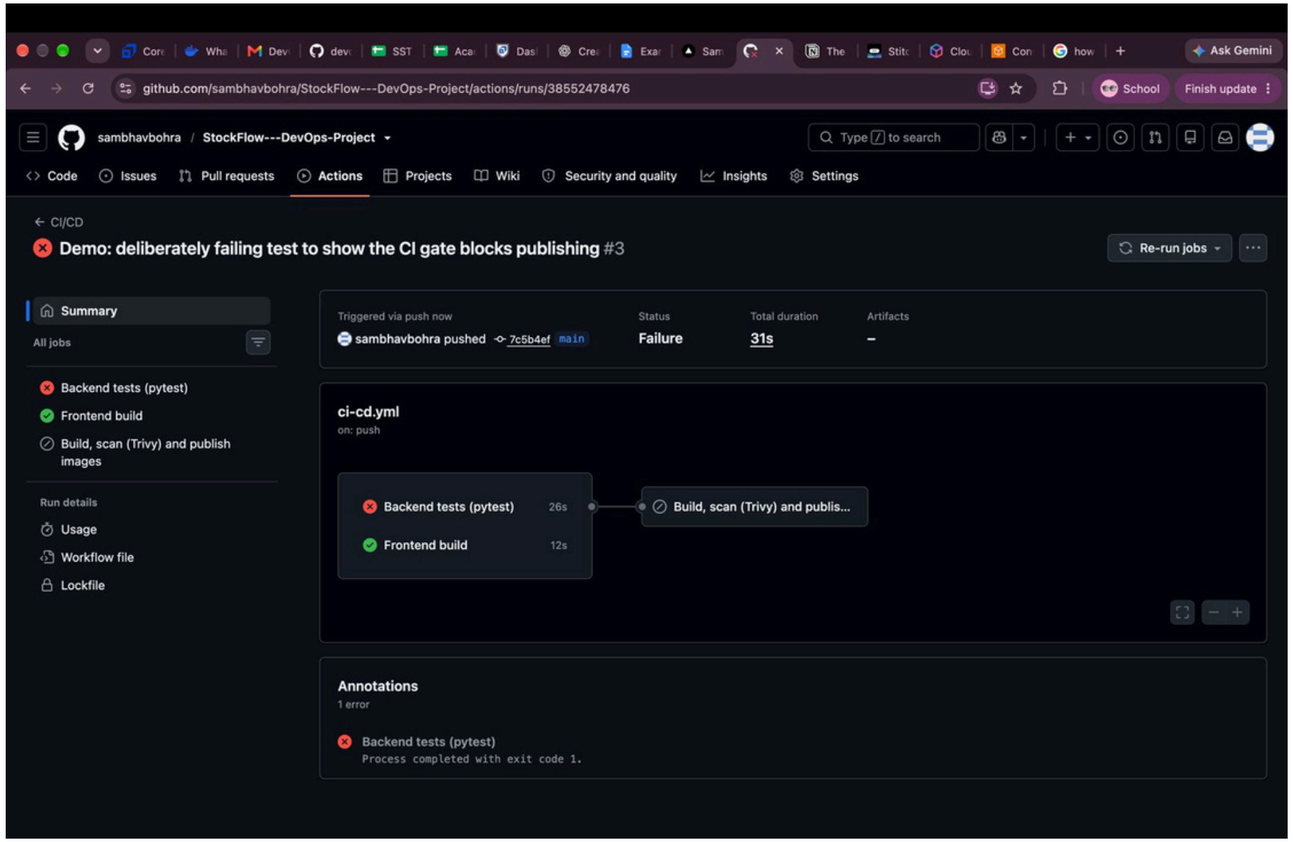
CI/CD pipeline

- On every push to main: pytest + frontend build → build images → Trivy → GHCR (commit-SHA tags) → Helm deploy
- A failing test blocks publishing (demonstrated red run)
- Deploy via GitHub OIDC: no AWS keys stored



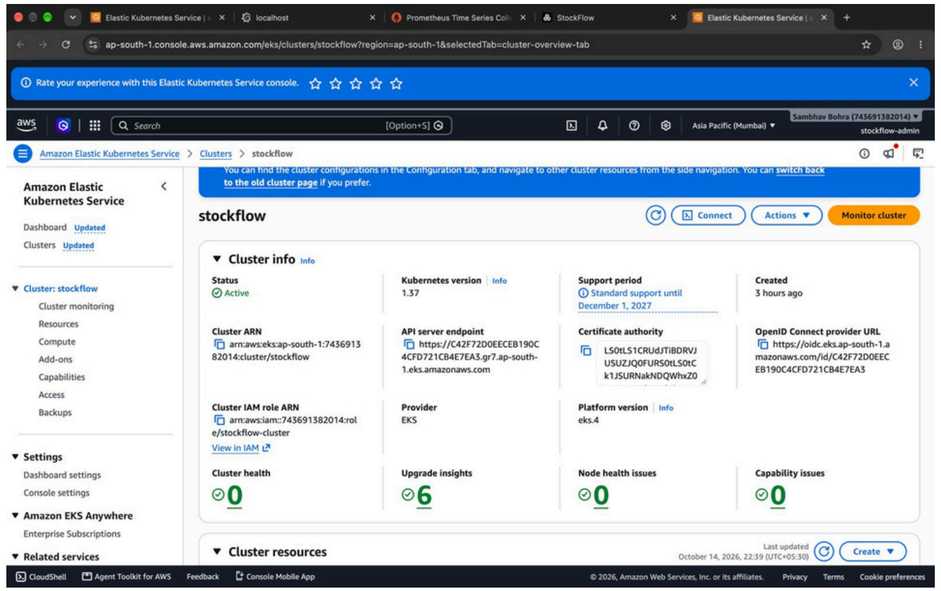
DevSecOps

- Trivy fails the build on fixable HIGH/CRITICAL CVEs; it found pip-bundled CVEs, fixed by removing pip from the runtime image
- Actions pinned to commit SHAs; Trivy installed with a verified checksum (after the 2026 trivy-action hijack)
- Non-root containers; secrets only in Kubernetes Secrets, never in Git



Infrastructure as code (Terraform)

- VPC with 2 public subnets (2 AZs), no NAT gateway
- EKS 1.37 (standard support pinned), managed Spot node group (c7i-flex.large ×2)
- Add-ons: VPC CNI, CoreDNS, kube-proxy, Pod Identity, EBS CSI, metrics-server
- GitHub OIDC deploy role scoped to the stockflow namespace




```
Plan: 28 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + cluster_name            = "stockflow"
  + configure_kubectl       = "aws eks update-kubeconfig --region ap-south-1 --name stockflow"
  + github_actions_role_arn = (known after apply)
  + public_subnet_ids       = [
      + (known after apply),
      + (known after apply),
    ]
  + region                  = "ap-south-1"
  + vpc_id                  = (known after apply)

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these
actions if you run "terraform apply" now.
sammybohra@SammyBohras-Mac terraform %
```

Kubernetes + Helm

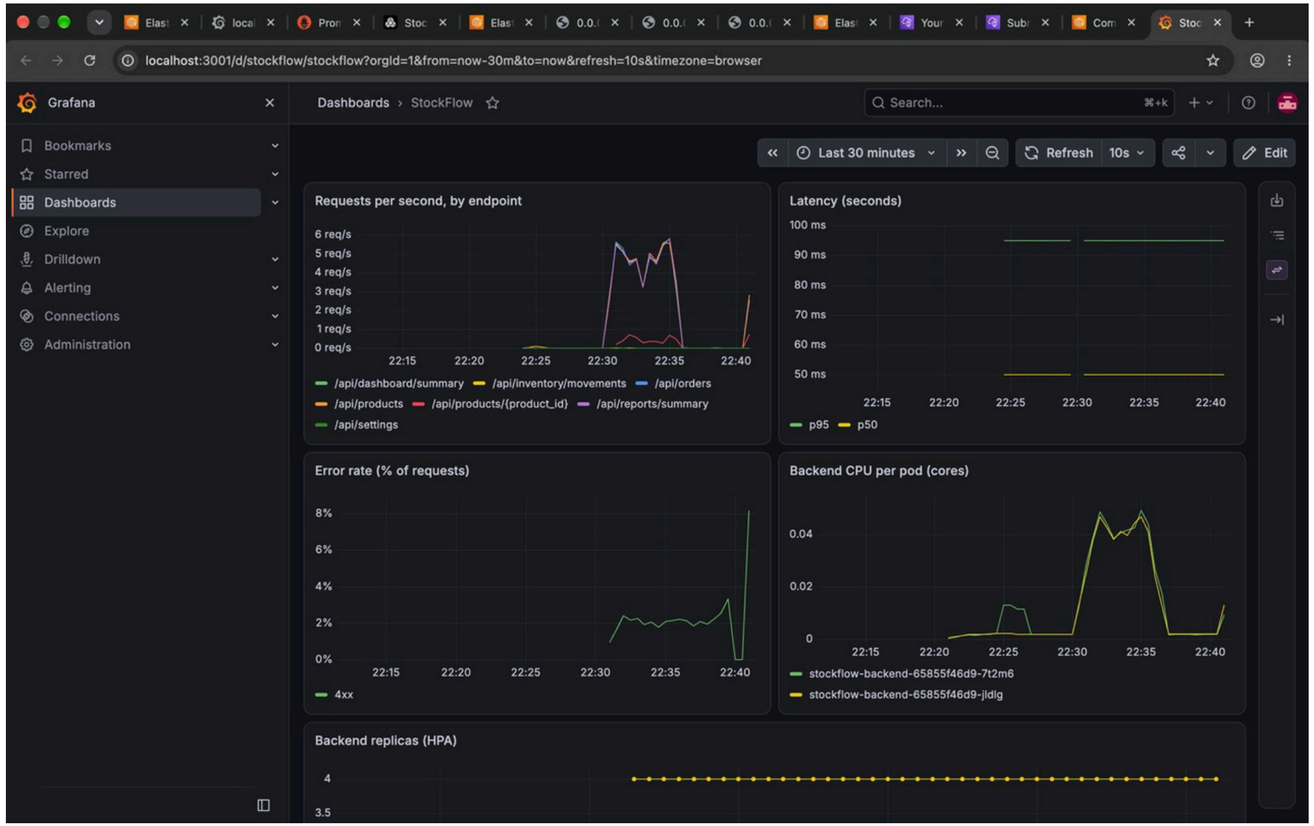
- Chart: Deployments (2 replicas, probes, limits), ClusterIP Services, Ingress (/ and /api), HPA (2–4 at 70% CPU), PostgreSQL StatefulSet + 1 GiB gp3 volume
- Zero-downtime rolling updates; image tag = commit SHA

```
sammybohra@SammyBohras-Mac StockFlow – DevOps Project % kubectl get pods -n stockflow -o wide
NAME                                READY   STATUS    RESTARTS   AGE     IP           NODE                                           NOMINATED NODE   READINESS GATES
stockflow-backend-65855f46d9-7t2m6    1/1     Running   0          9m35s   10.0.1.27    ip-10-0-1-57.ap-south-1.compute.internal   <none>           <none>
stockflow-backend-65855f46d9-jld1g    1/1     Running   0          9m35s   10.0.0.109   ip-10-0-0-88.ap-south-1.compute.internal   <none>           <none>
stockflow-frontend-64c7968b66-5rsmp   1/1     Running   0          9m35s   10.0.0.189   ip-10-0-0-88.ap-south-1.compute.internal   <none>           <none>
stockflow-frontend-64c7968b66-6r24q   1/1     Running   0          9m35s   10.0.1.100   ip-10-0-1-57.ap-south-1.compute.internal   <none>           <none>
stockflow-postgres-0                1/1     Running   0          9m35s   10.0.1.216   ip-10-0-1-57.ap-south-1.compute.internal   <none>           <none>
sammybohra@SammyBohras-Mac StockFlow – DevOps Project %
```

```
sammybohra@SammyBohras-Mac StockFlow – DevOps Project % helm list -A
NAME            NAMESPACE       REVISION        UPDATED                                 STATUS          CHART                                   APP VERSION
monitoring      monitoring      1               2026-10-14 22:20:01.242748 +0530 IST    deployed        kube-prometheus-stack-92.6.0    v0.94.1
stockflow       stockflow       1               2026-10-14 22:19:18.431497 +0530 IST    deployed        stockflow-0.1.0                 1.0.0
traefik         traefik         2               2026-10-14 22:23:30.27524 +0530 IST     deployed        traefik-41.7.1                  v3.7.14
sammybohra@SammyBohras-Mac StockFlow – DevOps Project %
```

Observability

- Backend /metrics → ServiceMonitor → Prometheus (targets UP) → Grafana dashboard
- Panels: requests/s, p50/p95 latency, error %, CPU per pod, HPA replicas



Real problems I solved

- Free-plan account rejected t3 instances → switched to Free-tier-eligible c7i-flex.large
- Account not allowed to create load balancers → Traefik NodePort + one security-group rule
- OIDC deploy denied → CloudTrail showed GitHub's new ID-based sub claim → fixed the trust policy
- Backend crash-looped before the DB was ready → init container waits for PostgreSQL
- Troubleshooting lab: broken image (ImagePullBackOff) and broken Service selector (no endpoints)

Results, cost & teardown

- All rubric modules M1–M10 covered, with evidence screenshots in screenshots/
- Cost: about $0.20/hour (Spot, no NAT, no load balancer), covered by credits; $2 budget alert
- Teardown: helm uninstall → terraform destroy → verify no leftovers

Thank you.